

Prompt:

Write a Python script to perform a linear regression analysis using the dataset located at "C:\Users\MatebookD16\Downloads\Lab03\lab report.xlsx".

The script must:

1. Load the data from the specified Excel file and set the independent variable (x) as Total Floor Area (m^2) and the dependent variable (y) as Total Construction Value (PHP '000).
2. Compute the least-squares regression line (slope and intercept), R^2 , and standard error dynamically from the dataset.
3. Calculate a prediction for a target floor area of 500,000 m^2 .
4. Generate a professional visualization using a **lavender/violet aesthetic theme** that includes:
 - A regression fit plot showing the data points and trend line.
 - A residuals plot showing the error distribution.
 - A results text block displaying the computed regression metrics and prediction value.